# CA Assignment 2

## Clustering Algorithms

| Assignment Number | 2 of (2) |
|---|---|
| Weighting | 15% |
| Assignment Circulated | 10.03.2025 |
| Deadline | 27.03.2025 |
| Submission Mode | Electronic Via Canvas |

1. You are provided with a dataset containing 1,760 data instances. Each line in the dataset file corresponds to one data instance. Your task is to cluster this dataset using any clustering algorithm of your choice. You may also apply any appropriate data preprocessing techniques (e.g., scaling, normalization, feature selection, etc.) before clustering. Your code should not involve any kind of cloud API call which will require subscription. (70)

**Submission Requirements** You must submit the following:

- **Silhouette Analysis Plot (10 marks)** Submit a silhouette coefficient plot (a .png or .jpg file) for values of $K$ ranging from 1 to 10. The x axis will have K values and y axis will have the silhouette coefficient values.
- **Cluster Labels File (10 marks)**
  - Submit a file named 'label.txt' that contains only a single column.
  - Each row of 'label.txt' must contain the cluster label assigned to the corresponding data instance.
  - The order of labels must exactly match the order of instances in the original dataset file.
  - Following is an example of a label file containing 10 data points and two different clusters.

    1<br>2<br>1<br>1<br>2<br>2<br>1<br>1<br>2<br>2
- **Code (30 marks)**
  - Submit all code used for preprocessing and clustering in a single zip file called cluster.zip. Preprocessing code should be saved in a file named 'preprocessing.py'. Clustering code should be saved in 'clustering.py'. If you are using any library which is not installed by default in a machine, then you should mention that in a 'requirements.txt' file. If you fail to do so then your marks will be deducted (5 marks).
  - The code should be well-structured, readable, and reproducible (5 marks).
  - The file clustering.py must be written in a way so that it can be executed on a new dataset with the same format as the one provided. When run, it should automatically generate a file

named label.txt containing the predicted cluster label for each data instance. We will just do 'python clustering.py data.txt' (20 marks).

- **Preprocessing Explanation (10 marks)** Submit a separate document (a .txt or .pdf file) explaining:
  - What preprocessing steps were applied
  - Why they were applied
  - How they may influence clustering performance
- **Clustering Results Explanation (10 marks)** Submit a document explaining:
  - The clustering algorithm used
  - What you believe is the optimal value of $K$ and why (based on silhouette analysis or other reasoning)

2. Compute the confusion matrix, macro-averaged Precision, Recall, and F-score for the clustering shown in Figure 1. (20)

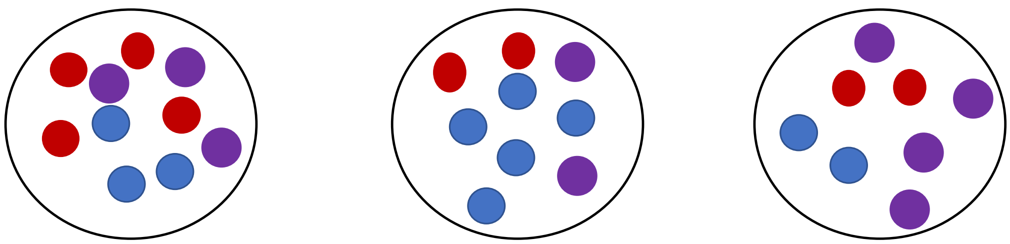

Figure 1: Outcome of a Clustering Algorithm

**Submission Requirements** You must submit a document (.txt or .pdf file) showing the computation of the different metrics. No code needs to be submitted for this exercise.

3. For the same clusters as in Figure 1, compute B-CUBED Precision, Recall, and F-score. (10)

**Submission Requirements** You must submit a document (.txt or .pdf file) showing the computation of the different metrics. No code needs to be submitted for this exercise.

## Important Notes

1. You are free to use any clustering algorithm or preprocessing approach.
2. Filenames should be exactly the same as instructed above.
3. Programs that do not run will result in a mark of zero!
4. Compilation error in your code will give zero marks. We will do 'python clustering.py' then it should work.
5. Your code should be as clear as possible and should contain only the functionality needed to answer the questions. Provide as much comments as needed to make sure that the logic of the code is clear enough to a marker. Marks may be deducted if the code is obscure, implements unnecessary functionality, or is overly complicated.

6. If you use module random to make some random actions, use a fixed seed value so that your program always produces the same output.
7. Your submission should be your own work. Do not copy or share! Make sure that you clearly understand the severity of penalties for academic misconduct (`https://www.liverpool.ac.uk/media/livacuk/tqsd/code-of-practice-on-assessment/appendix_L_cop_assess.pdf`).